KOLHAPUR INSTITUTE OF TECHNOLOGY'S
COLLEGE OF ENGINEERING KOLHAPUR (EMPOWERED AUTONOMOUS)
(AFFILIATED TO SHIVAJI UNIVERSITY, KOLHAPUR)

Kolhapur Institute of Technology's College of Engineering (Empowered Autonomous) Exam. Cell

Final Year B. Tech (Computer Science and Engineering)
(Semester - 7)
END SEMESTER EXAMINATION, DEC-2025

Course Code : UCSC0702

Course Name : Advanced Database Systems

Day and Date : Thursday 18th Dec, 2025 **PRN :** 2223000649

Time : 02:00 PM to 04:00 PM **Max Marks : 50.00**

Instructions:

IMP: Verify that you have received the question paper with the correct course, code, branch, etc.

i. All questions are compulsory.
ii. The figure to the right indicates full marks.
iii. Assume suitable data wherever necessary.

	B. L.	CO's	Marks
Q. 1 Attempt any two			
A. What is table inheritance in SQL? Write queries for: i) Creating a user-defined type Employee having attributes fullname of user-defined type FullName, contact of user-defined type ContactInfo, and hireDate of type date. ii) Create a table staff of user-defined type Employee. iii) Define tables engineers (qualification, project) and managers (salary, teamSize) as sub-tables of staff.	1,3	CO1	8.00
B. Illustrate different partitioning techniques in parallel database systems with its advantages and disadvantages.	2	CO2	8.00
C. i)Explain cursor in SQL with syntax . ii)A company maintains an Employee table that stores information about its employees, including their IDs, names, departments, and salaries. Write cursor to retrieve all employees with a salary below Rs.50,000	3,2	CO3	8.00
Q. 2 Attempt any two			
A. Describe column- family database? Write queries for following the cassandra database. 1. Insert a record in table student_course(stud_rollnumber int , stud_name text, stud_branch text ,stud_PRN int , primary key(stud_PRN) . 2. Use collection datatype for adding new column in student_course table(such as stud_email,stud_subject,stud_mobilennumber). 3. update student record.	3,2	CO4	8.00
B. Outline some popular key-value store databases. Discuss any one key-value Database with its properties.	2	CO4	8.00
C. What is Business intelligence (BI)? Explain the basic BI Architectural Components.	2	CO5	8.00

	B. L.	CO's	Marks
Q. 3 Attempt any three			
A. Illustrate the ACID and BASE properties in perspective of NoSQL.	2	CO4	6.00
B. Explain following things with respect to graph database: i) Advantages of Graph Database ii) Applications of Graph Database iii) Features of Graph Database	2	CO4	6.00
C. What Is a Document Database? Describe following basic operations on Document Databases Inserting Deleting Updating Retrieving	1	CO4	6.00
D. Outline Datawarehouse architecture briefly.	2	CO5	6.00